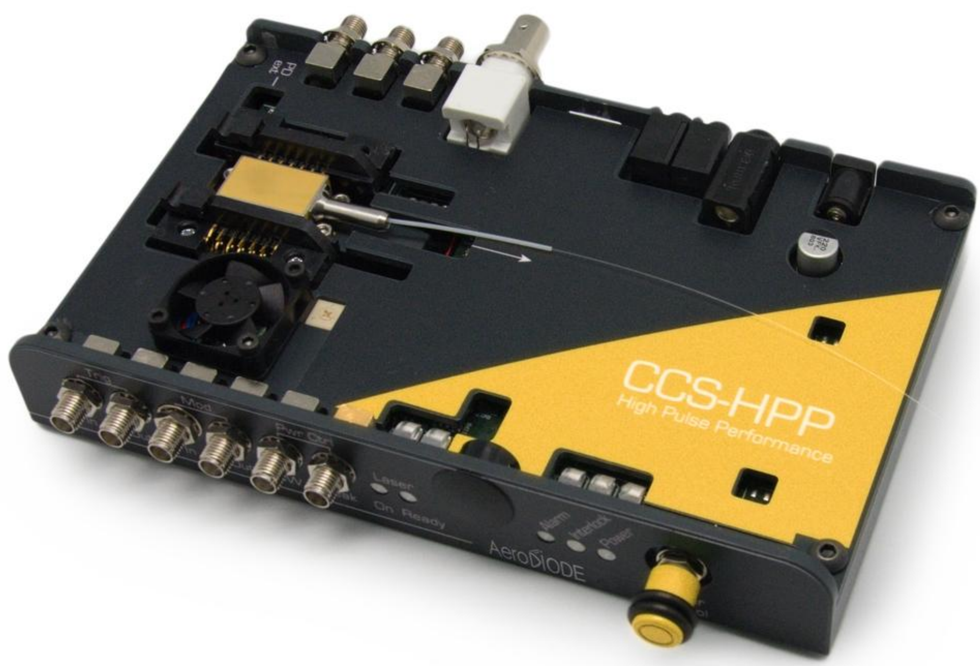
User Manual

Cool & Control Series

CCS-HPP



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Contents

Revision Sheet

Release No.	Date	Author	Revision Description
V1.0	22/07/2020	AMU	First version
V1.1	19/11/2021	SER	Minor modification (picture laser diode pinning updated)
V1.2	11/07/2022	AKU	ControlSoftware update
V1.3	13/06/2023	AMU	Added environmental friendliness chapter and clarification about diode and TEC statuses in the GUI
V1.4	21/06/2024	LFO	Added key-control option and AeroDIODE End of Life Policy
V1.5	11/03/2025	LFO	Added AeroDIODE WEEE number, better description of the switch.

Disclaimer

Information in this document is subject to change without notice.

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1. General information

Please read this manual carefully, it describes the hazard the user might be exposed to while using the product. It also explains in details how to use the product in the safest and more efficient way possible.

The safety of any system incorporating the product is the responsibility of the assembler of the system.

Any action taken by the user that is not clearly described in this user manual might present a risk and is the sole responsibility of said user.

This product is to be used in laboratory or industrial tasks, and only by personnel who have followed a training in laser hazard.

1.1 Definitions

Warning
A warning is given for potentially dangerous situation for people which cause them harm or lead to death.

Caution
A caution is advised when dealing with hazardous situations, tasks or objects, to prevent Material damage or failure.

Note
A note is a complementary piece of advice that must be acknowledged by the user.

1.2 General Warning & Caution

Warning
Do not try to open or remove the cover of the CCS-HPP module.

Warning
Any software settings or hardware tinkering that is not described in this user manual or in the usage recommendation may put the user or its environment at risk.

Warning
The maintenance and servicing of the CCS-HPP should not be executed by the end user: only AeroDIODE is able to maintain the CCS-HPP.

Warning
The compatible laser diodes used with the CCS-HPP can deliver up to several Watts of coherent LASER radiation. Always wear protective goggles and observe the safety instructions provided by the laser diode supplier when using the CCS-HPP driver with your laser diode.

Caution
Avoid all chocs and strains when handling the CCS-HPP.

Caution
Handle the fiber-optics cable with care as it is fragile. Do not bend or pinch it.

Note
Only use the genuine power supply, and the supplied USB cable.

2. Safety Instructions

2.1. Wiring

Caution

- Please first connect the input pin to the board and then plug the DC Power Supply to the power grid.
- Use caution when connecting the Power Supply.
- Protect the power cord from being walked on or pinched particularly at plugs, convenience receptacles, and the point where they exit from the CCS-HPP module.
- Connect the ground completely. Electric shock may occur if the ground is not connected correctly.
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2.2. Operating Environment

Caution

- Do not install near any heat sources such as radiators, heat registers, stoves, or other equipment (including amplifiers) that produce heat.
- To reduce the risk of fire or electric shock, do not expose the CCS-HPP to rain or moisture.

Warning

Not following the safety recommendations and the caution mentioned above can lead to eye damage.

2.3. Contact

If you have any question about the CCS-HPP module, please contact AeroDIODE.

3. AeroDIODE “End of Life” Policy

This product complies with the WEEE (Waste Electrical and Electronic Equipment) directive of the European Community and the corresponding national laws.

The affixed label indicates that you must not dispose of the product in a litter bin or in domestic household waste. This product must be handed to a company specialized in waste recovery.

AeroDIODE has an obligation to ensure the appropriate steps are taken for the recovery and recycling of WEEE from its business customers. Once the product has reached the end of its life, it must be returned to AeroDIODE or your local sales partners. To meet this obligation we ask customers to contact AeroDIODE or local sales partners (www.aerodiode.com) to organize the take-back. The return of end of life products is free of charge.

Eligible products of the take-back service are marked with the “crossed out wheelie bin” symbol, sold and currently hold by a company or institute within the EC and not disassembled.

WEEE Number (FRANCE) : FR403931_05SQKS

The intent of the WEEE directive is to enforce the recycling of WEEE. A controlled recycling of end of life products will thereby avoid negative impacts on the environment.

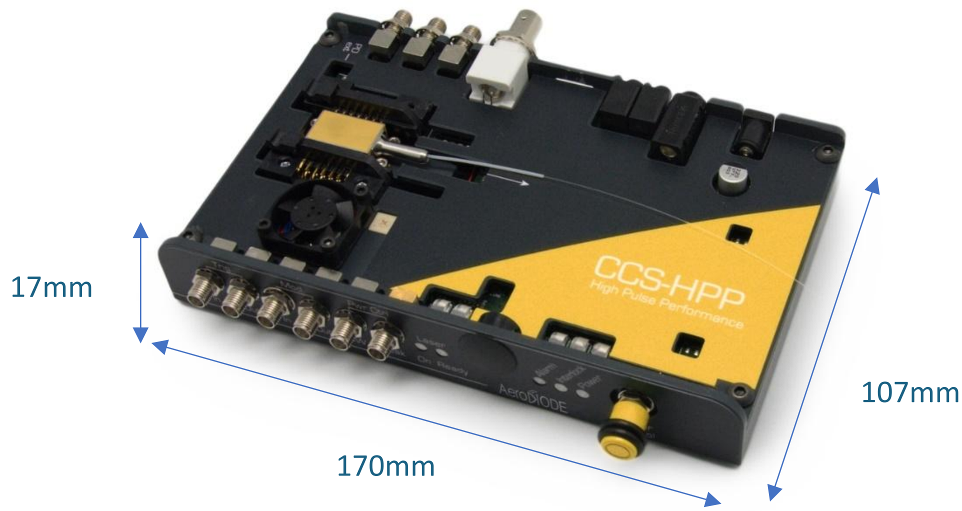
Thank you for your interest in ensuring your AeroDIODE product is handled effectively at the end of its useful life.

4. Package Content

The CCS-HPP package comes with:

- 1 CCS-HPP HPP
- 1 DC Power Supply (+24V DC / 2A)
- 1 USB type A to type B cable
- 1 USB Key with "AeroDIODE Control Software Suite", test report, troubleshooting guide & User manual in .pdf

5. Product overview



Physical characteristics		
Length	170	mm
Width (edge to edge)	107	mm
Width (Connector to connector)	141	mm
Height (top plate)	17	mm
Height with fins	30	mm
Weight	464	g

5.1. Front view of the product

1. **Trigger (SMA)**
   a. In **(SMA input, 50 Ohm impedance)**: Input pulse command/trigger. SMA TTL/LVTTL input voltage connector. The signal must be between 0 and +3.3V.
   Max bandwidth @-3 dB : < 500 MHz.
   b. Out **(SMA output, 50 Ohm impedance)**: Numerical output of the laser pulse (LVTTL). Max bandwidth @-3 dB : < 250 MHz.

2. **Modulation (SMA)**
   a. In **(SMA input, 10 kOhm impedance)** : Analog voltage input (±10 V) to modulate the laser power. Max bandwidth @-3 dB : < 3 MHz.
   b. Out **(SMA output, 10 kOhm impedance, 2.5 V offset, 0.5 V/A (2A version) or 2 V/A (500 mA version)** : Analog voltage output to view the laser external or internal modulation.

3. **Power control (SMA input with 10 kOhm impedance)**
   a. CW : Laser continuous power adjustment (analog input). Apply a voltage between 0 and +5V for power adjustment. Max bandwidth @-3 dB : < 0.1 Hz.
   b. Pulse : Laser peak power adjustment (analog input). Apply a voltage between 0 and +5V for power adjustment. Max bandwidth @-3 dB : < 300 Hz.

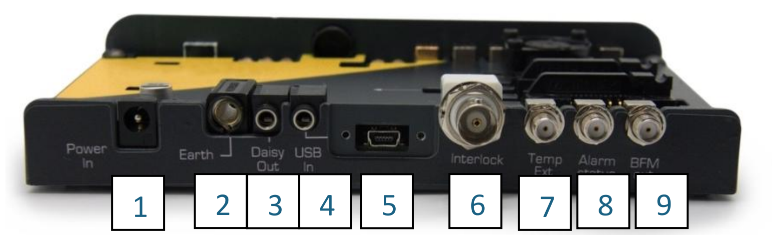
4. **Laser status (blue LED)**
   a. On: The laser is ON in the software.
   b. Ready: The laser is ready for emission (no alarm, interlock closed)

5. **Safety key control (optional)**
   Turn the key to enable/disable the laser "Ready" status. The key must be turned OFF/ON in case of an alarm to allow the laser emission again. This key is removable and the laser could not be operable when the key is removed.

6. **Status (red/green LED)**
   a. Alarm: Red LED indicates that the product detects an alarm (temperature, power, …). Please see in the software interface the type of alarm.
   b. Interlock: Red LED indicates that the product is not ready for laser emission. The BNC interlock on the rear is open, the key (if present) is not in the right position or an alarm is activated
   c. Power: Green LED indicates that the product is powered

7. **Signal peak or CW power adjustment (manual knob)**
   Knob for peak or CW power adjustment. In order to use the knob, the current source should be selected to “POT” in the software (see §8.3).

5.2. Rear view of the device



1. DC power input (+24 VDC)

Input connector for DC power. Use only the provided DC power supply.

2. Earth connection of the product (4mm test patch cord)

Input the earth into the jack connector to ground the product. This connection is not required for normal operation but can be used for maintenance purpose by AeroDIODE.

3. Daisy output (3.5mm jack)

It is used to chain multiple AeroDIODE products with different addresses. Use jack input/output to serial chain products with a single USB connection to the computer

4. USB input (3.5mm jack)

Use a “FTDI 3,3V RS232” USB/jack (jack/jack if products are chained) to connect the product to a computer

5. USB input (mini-USB B Type)

Connect the product to the computer with a USB/USB cable

6. Interlock (BNC)

Input connector for interlock control. It must be shunt in order to have a laser emission.

7. **External temperature (not used for this application) (SMA input with 10 kOhm *impedance)***

Input a voltage between 0 V and +5 V to change the laser internal temperature.
Max bandwidth @-3 dB : < 300 Hz.

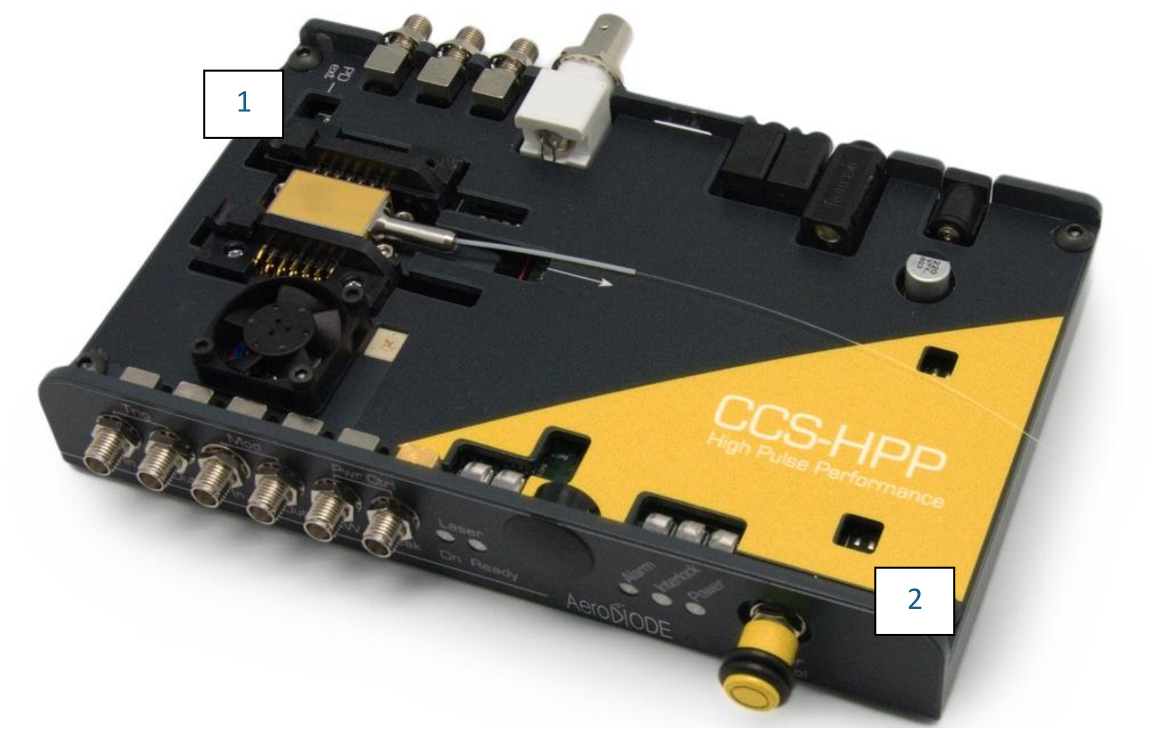
8. Alarm monitor (SMA output, 100 Ohm impedance)

Output a signal for alarm status (Power or Temperature) monitor. The signal is a direct LVTTL, so a +3.3 V output voltage level corresponds to alarm ON.

9. BMF output (SMA with 50 Ohm)

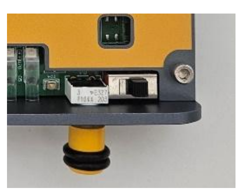
Back facet monitor output. This analog signal gives an electrical signal that is a view of the laser optical output. Max bandwidth @-3 dB : < 1 GHz.

5.3. Top view of the device



1. **PD_Ext In**
   Molex connector to connect an external photodiode

2. **Switch**



Left		Right
24V forced	**Neutral**	**Adjustable voltage**
Ideal for pulse operation performances	Prevent lasing	For continuous operation and standard pulse operation*

*In CW operation, if the adjustable voltage is set too high, the MOS CW alarm might get triggered, in that case the product will no longer emit.

6. Installation

6.1. Butterfly Laser diode installation

The CCS-HPP driver is made to drive butterfly laser diodes (14 pins) with internal monitoring photodiode and thermoelectric cooler. The laser diode output fiber must go through the CCS-HPP package.

Note
Use a torque screwdriver for the diode and please refer to the maximum allowed torque given in the diode specification (usually around 15 to 18cNm).

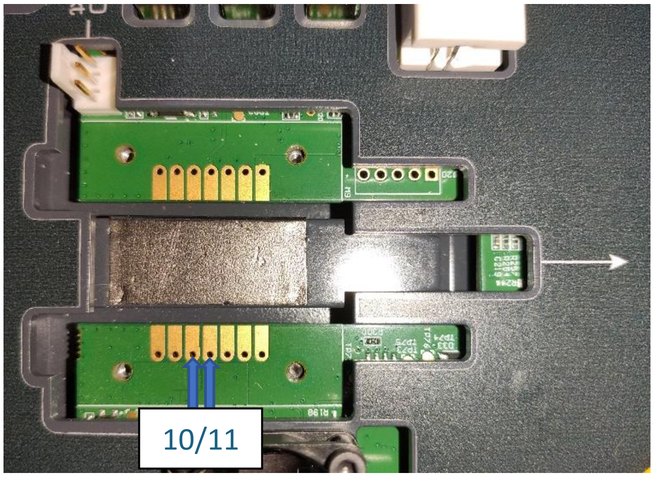
CCS-HPP must be adapted to the type of butterfly laser diode : Standard CCS-HPP are adapted for type-pinning laser diode (see §11).

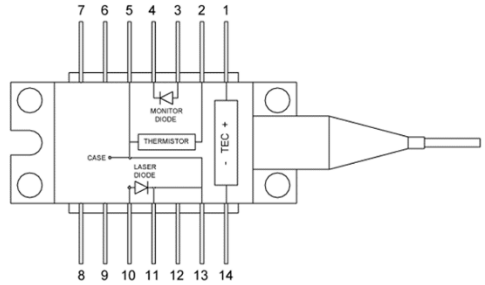
Fiber output direction
For SOA see bellow

- Non-butterfly laser diode installation

In case a standard non butterfly laser diode is used (like TO form factor), It is relevant to order a CCS-HPP without the Butterfly mounting sockets. This allows to solder the laser diode directly on the Board soldering pads.

The anode must be soldered on pin 10 and the cathode on pin 11 (see the figure bellow showing pin 10 and 11 by analogy with a butterfly form factor pinning)

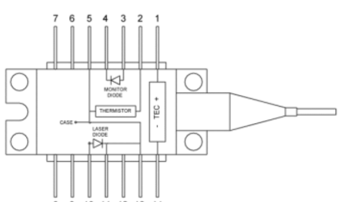


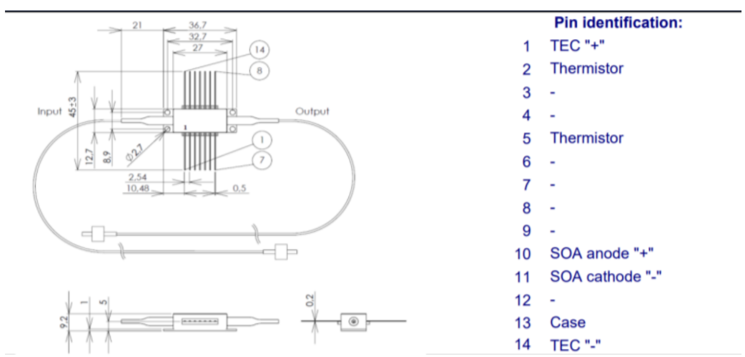


Note that the CCS-HPP can be held vertically so that the laser diode emits horizontally.

6.2. SOA installation

In case a SOA is used, make sure the SOA is mounted in the correct direction. Refer to the pin of the SOA (below, right hand side) and compare with a standard Type 1 laser diode (below, left hand side) :





In the example above, one can see that the fiber output of the laser diode above (see CCS-HPP picture with fiber direction) corresponds to the input of a SOA.

- Software installation

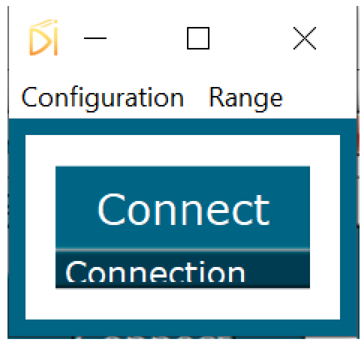
The CCS-HPP USB specific cable should be unplugged during the installation.
Double-click on setup.exe to run the installer. The control software will be installed, as well as the driver for the USB cable. A computer restart may be required to complete the installation.

7. Getting started

- When the software is installed, plug the USB cable into a USB port of your computer.
- Next, plug the USB cable in the USB port of the CCS-HPP.
- Plug-in the CCS-HPP power supply to turn on the CCS-HPP laser diode driver

Click on the item "AeroDIODE Control Software" located in the Start Menu to run the CCS-HPP control software.

A window will appear:



Click on Connect to start the CCS-HPP hardware detection. The software will automatically detect any USB-connected CCS-HPP.

A new window will appear for each CCS-HPP.

The window for laser controls is shown below:

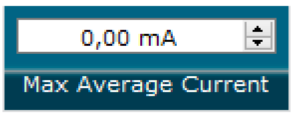
The window is divided in six parts:

1. **Maximum levels** : three numerical boxes used to change the maximum allowed current (power) for the diode.
2. Triggers in the **Control** section are used to select the operating mode.
3. The **Configurations** section is for advanced options.
4. Numerical boxes in the **Settings** section are used to configure the pulse width, amplitude and repetition rate and laser diode temperature as well as the compliance voltage.
5. The **Measurements section** contains read only numerical boxes used to monitor the module.
6. The **Alarms** part shows green/red indicators for various status and a button to open a complete measure window.

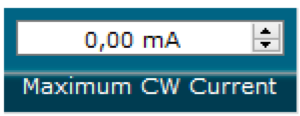
8. User interface description

8.1. Maximum levels

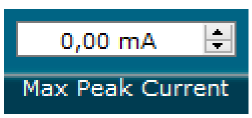
Use those numerical boxes to configure the maximum current (power). The values which can be entered in these numerical boxes are factory limited. The limits depend on various parameters such as the wavelength and the maximum allowed optical power.



The average current is a hardware protection which estimates the average current in continuous but also pulsed operating mode. In case the duty cycle and peak power are too high this security will protect the laser

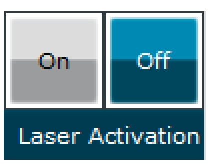


The absolute maximum continuous current in mA can be reduced here to protect the laser diode.

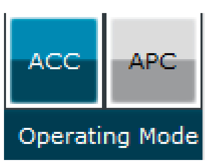


The absolute maximum peak current in mA is factory configured in order to protect the laser but can be reduced here. It can be useful as the current resolution will be better in case of lower current

8.2. Control : control laser emission and operating mode

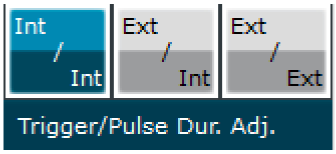


Enables/Disables the laser emission

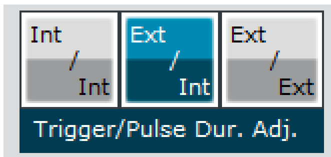


ACC (automatic current control) mode must be used in order to drive the laser with a fixed current.
APC (automatic power control) mode is used for driving the laser based on a fixed power (Not used for this application in pulsed regime).

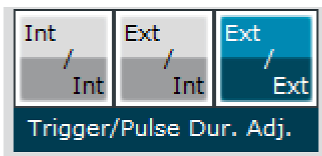
Select between three pulse operating mode:



Int/Int: the frequency and pulse width are configured from the software (internal trigger and internal pulse width adjustment, all software)



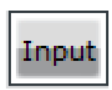
Ext/Int: the signal is externally triggered by a digital signal on the SMA input "Trig In" (control number 1 on §5.1) and the pulse width is configured from the software (external trigger, internal pulse width adjustment)



Ext/Ext: the output optical signal is generated from the digital input signal based on the SMA "Trig In" (control number 1 on 0). The optical signal will be a copy of the digital input signal (external trigger, external pulse width adjustment)

8.3. Configuration

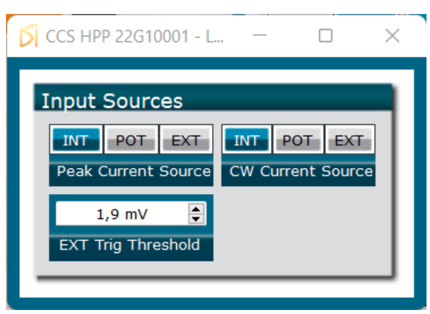
Click on the Input button in the configuration part to open a new window for advanced configuration settings



In this window the user can change the source for adjusting the current from:

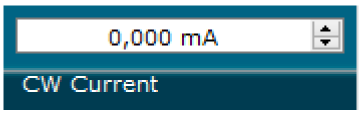
- INT: internal (software)
- POT: external adjustment with the manual knob (see control number 6 on §0)
- EXT: external adjustment with the SMA input (see control number 3 on §0)

The user can also change the trigger level threshold for the SMA trigger input (see control number 1 on §0.

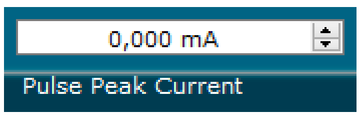


8.4. Settings: configure power and pulses

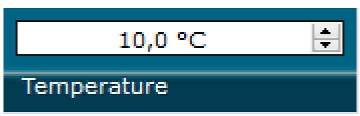
The values which can be entered in these numerical boxes are limited by maximum level (described in §8.1).
The user could apply a continuous power and a pulse power at both times. The continuous power applied will not change the peak power, it will act like a DC bias power (the only limitation will come from the maximum average current allowed).



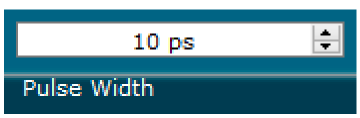
Continuous current in mA. This item can only be modified when the internal CW current source is selected



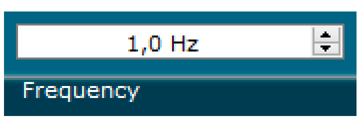
Peak current in mA. This item can only be modified when the internal Peak current source is selected



This item configures the internal temperature of the laser diode (on the range 10°C/50°C). The typical value is 25°C and it is not recommended to change it for this application



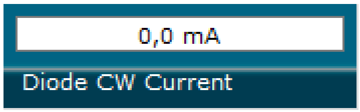
Internal pulse width adjustment up to 500ns. This item can only be modified with an internal pulse width adjustment. The pulse width could be higher than 500ns when using the Ext/Ext mode



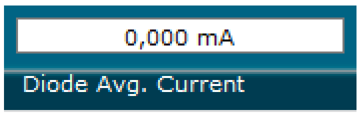
Internal frequency (repetition rate) adjustment up to 250MHz. This item can only be modified when the internal trigger is active. The frequency could be higher than 250MHz when using the Ext/Int or Ext/Ext mode

8.5. Measures: read only product monitoring

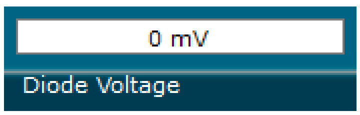
The numerical boxes below are read-only and can't be modified by the user.



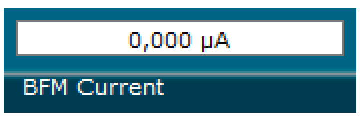
Hardware measurement of continuous current (when a non-null continuous current is applied)



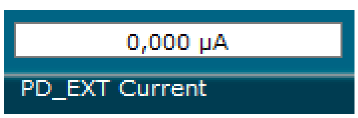
Hardware measurement of average current which is the addition of continuous current and pulsed current



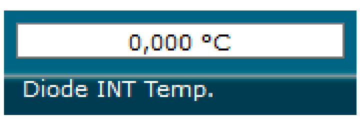
Hardware measurements of the voltage of the diode (Beware : when the **interlock** is **not shunted**, the reading corresponds to the driving voltage, **not the diode voltage**)



Hardware measurement of average optical power returned by the laser's internal photodiode. The value is not calibrated and depends on the laser source

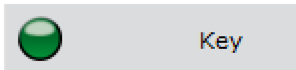


Optional. If present, hardware measurement of average optical power returned by an integrated photodiode

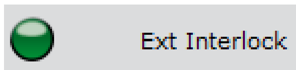


Hardware measurement of internal temperature of the laser. It should be near the temperature setpoint.

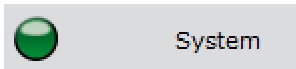
8.6. Alarms



(Legacy indicator) Red light indicates that the key is on the OFF position *(optional)*

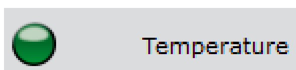


(Optional) Red light indicates that an external interlock is detected

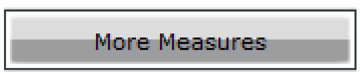


Red light indicates that the product returns an internal alarm

Red light indicates that the product is limited by the maximum average current (power). When alarm is ON the laser is temporally OFF until the average current goes below the maximum



Red light indicates the temperature is too far from the setpoint (+/-1,5°C). When alarm is ON the laser is temporally OFF until the consign temperature is reached



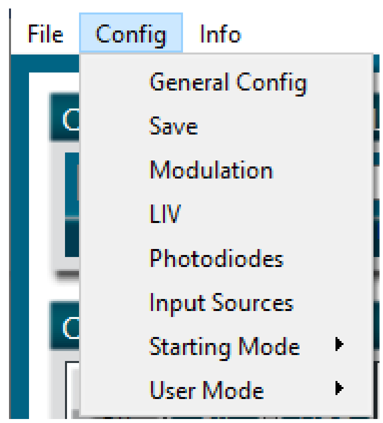
This button will open a new window to have more measures. It should be useful in case of technical support

9. Menu bar

The menu bar contains three main menus:

- **File:**
  - **Load parameters**: *prevision (does nothing)*
  - **Save parameters**: *prevision (does nothing)*
  - **Exit:** Close the window. The laser state can be either active or inactive after the software is stopped.

- **Config:**
  - **General Config**: *reserved for AeroDIODE only under password*
  - **Save**: save all the current operating parameters into the product internal memory. These parameters will be saved as default parameters in the flash memory and will be retrieved by the product at the next power on (even if the product is not connected to the PC).
  - **Modulation**: *give access to the modulation section*
  - **LIV**: *not used for this product*
  - **Photodiodes**: *not used for this product*
  - **Input source**: *same as described in §8.3*
  - **Starting Mode** can be used to change the way the product starts
    - **Previous OEM:** the product starts with the same parameters that were previously set. **So if the laser activation is set to ON and parameters saved into memory, the lasing will be effective after a shutdown.**
    - **Previous:** (default) the product starts with the same parameters that were previously set except the laser activation which is forced to OFF
    - Zero current: with the same parameters that were previously set except the current value that is set to zero and laser activation forced to OFF
    - Low Alarm with the same parameters that were previously set except the maximum values that are set to a minimum, the current value that is set to zero and laser activation forced to OFF
    - Central activation: *reserved, not used for this product*
    - MMD activation: *reserved, not used for this product*
  - **User Mode**: *reserved*



- **Info:** Displays information about the current version and the internal parameters and open the full measures window

10. Technical Specifications

10.1. General Data

Length	170	mm
Width (edge to edge)	107	mm
Width (Connector to connector)	141	mm
Height (top plate)	17	mm
Height with fins	30	mm
Weight	464	g
Power connector (Jack, positive tip)	5,5	mm
Power supply (DC)	24V / 2A	
Safety Features	Interlock	
	Over Temperature Protection	
	Laser Current Limit	

10.2. Detailed data

CCS-HPP	Min	Max	Resolution	Impedance	Bandwidth
Operating temperature	-15°C	+40°C			
Storage temperature	-25°C	+70°C			
Operating Altitude	_	2000m			
Output current for CW mode	0 A	2000 mA	12 bits		
Output current for pulse mode	0 A	4000 mA	12 bits		
Pulse width (for internal generation)	0,2 ns	510 ns (1275ns on request)	For 0 to 10ns : 10ps For > 10ns : 2 ns		
Frequency	1 Hz	250 MHz	1 Hz		
Typical delays between trigger EXT and optical pulse	For Ext/Int Mode : • 0 to 10ns pulse duration : 70ns typ. • > 10ns pulse duration : 85ns typ. For Ext/Ext Mode : 30ns typ.				
Laser Diode Temperature	15°C	60°C	16 bi		
Trigger In level	(0 V)	TTL (5 V)		50 Ohm	250 MHz
Sync Out	LVTTL			50 Ohm	250 MHz
BNC peak power adjustment	0 V	5 V		47 KOhm	15 Hz
Alarms	0 V (Not active)	3.3 V (Active)		1 KOhm	

APENDIX

11. Supported laser diodes

11.1. Type 1 Butterfly laser : standard CCS-HPP

Standard CCS-HPP are well adapted for Type 1 Butterfly laser diodes :

N°	Description	N°	Description
1	TEC (+)	14	TEC (-)
2	Thermistor (TS)	13	Case Ground
3	PD node (-)	12	N/C
4	PD Cathode (+)	11	Laser Diode Cathode (-)
5	Thermistor (TS)	10	Laser Diode Anode (+)
6	N/C	9	N/C
7	N/C	8	N/C